

RSNA 2024 Lumbar Spine Degenerative Classification

赛后讲解

主讲人：H老师

赛题背景

医学图像多标签多分类 - 腰椎磁共振成像 (MRI)

这次比赛的目标是创建可以帮助检测和分类退行性脊柱疾病的模型，使用的是腰椎的磁共振图像。参赛者将开发模拟放射科医生在诊断脊柱疾病时表现的模型。

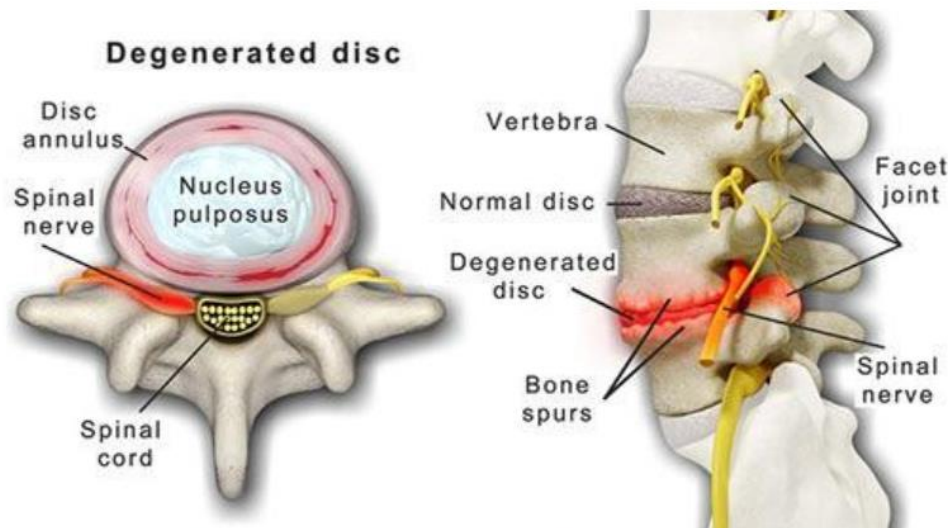
根据世界卫生组织的统计，腰背痛是全球致残的主要原因，2020年影响了6.19亿人。大多数人一生中的某个时刻会经历腰背痛，且随着年龄的增长，发生频率会增加。疼痛和活动受限通常是脊柱病的症状，这是一组退行性脊柱疾病，包括椎间盘退化和随后的脊柱管（脊柱狭窄）、副关节窝或神经孔的狭窄，伴有神经压迫或刺激的情况。

磁共振成像 (MRI) 提供了腰椎椎体、椎间盘和神经的详细视图，使放射科医生能够评估这些病症的存在和严重程度。准确的诊断和分级有助于指导治疗和潜在的手术，以缓解背痛，并改善患者的整体健康和生活质量。

挑战将重点放在五种腰椎退行性疾病的分类上：左侧神经孔狭窄、右侧神经孔狭窄、左侧副关节狭窄、右侧副关节狭窄和脊柱管狭窄。对于数据集中每个影像研究，提供了五种病症的严重程度评分（正常/轻度、中度或重度），涵盖了椎间盘水平 **L1/L2、L2/L3、L3/L4、L4/L5和L5/S1**。

竞赛概览

Treatment of Degenerative Spine Disease:

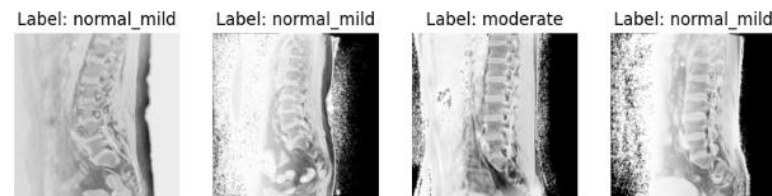


The treatment of degenerative spine diseases is not uniform and depends on the severity and type of condition. In many cases, surgery is not required and non-surgical treatment options can be used. Such treatments involve:

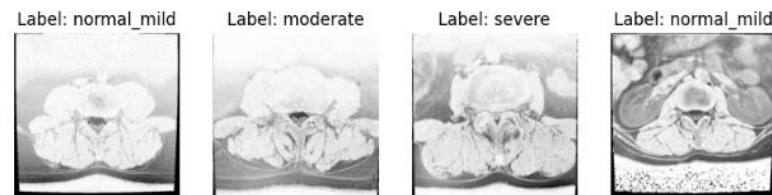
Medication Exercises to improve flexibility and muscle strength Braces To alleviate pain, doctors may prescribe pain medicine or steroids administered in a minimally invasive manner, usually using an epidural injection. In cases involving spinal stenosis or herniated discs, surgery may be required.

<https://www.kaggle.com/code/satyaprakashshukl/rsna-lumbar-spine-analysis>
<https://www.kaggle.com/code/shubhamcodez/rsna-efficientnet-starter-notebook>

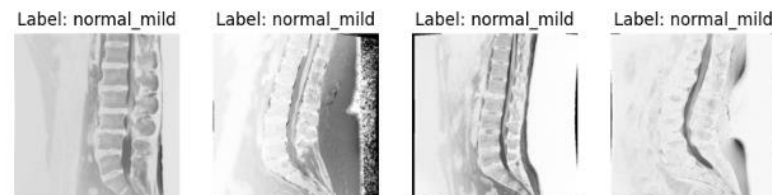
Visualizing Sagittal T1 samples



Visualizing Axial T2 samples



Visualizing Sagittal T2/STIR samples



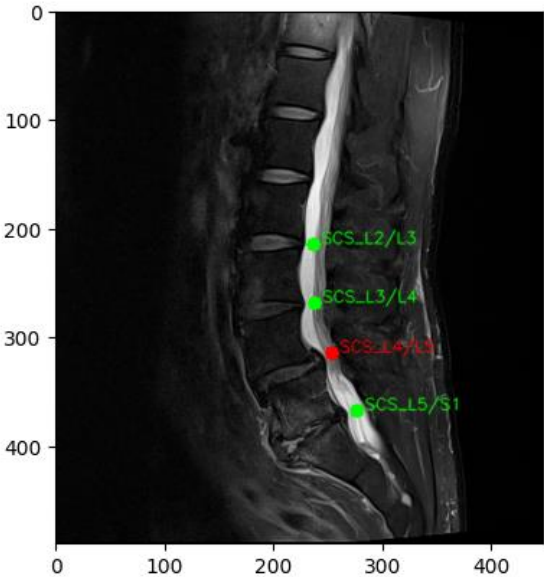
数据概览

- 1. 格式: [train/test]_images/[study_id]/[series_id]/[instance_number].dcm
训练集/测试集病例图像数据，由dicom格式给出。训练集有1974个病例，每个病例有N张扫描切片图像。
- 2. train.csv: 训练集标签
- 3. train_label_coordinates.csv: 训练集图像坐标信息表格数据。
- 4. [train/test]_series_descriptions.csv: 训练集/测试集图集序列信息。

```
In [3]: label_coordinates_df.head()
```

Out[3]:

	study_id	series_id	instance_number	condition	level	x	y
0	4003253	702807833	8	Spinal Canal Stenosis	L1/L2	322.831858	227.964602
1	4003253	702807833	8	Spinal Canal Stenosis	L2/L3	320.571429	295.714286
2	4003253	702807833	8	Spinal Canal Stenosis	L3/L4	323.030303	371.818182
3	4003253	702807833	8	Spinal Canal Stenosis	L4/L5	335.292035	427.327434
4	4003253	702807833	8	Spinal Canal Stenosis	L5/S1	353.415929	483.964602



数据概览

标签种类：

- Normal/Mild
- Moderate
- Severe

病症种类：

- Spinal Canal Stenosis 椎管狭窄
- Subarticular Stenosis 椎间孔狭窄（左右）
- Neural Foraminal Narrowing 神经根管狭窄（左右）

级别种类：

L1/L2、L2/L3、L3/L4、L4/L5、L5/S1

序列（视角）种类：

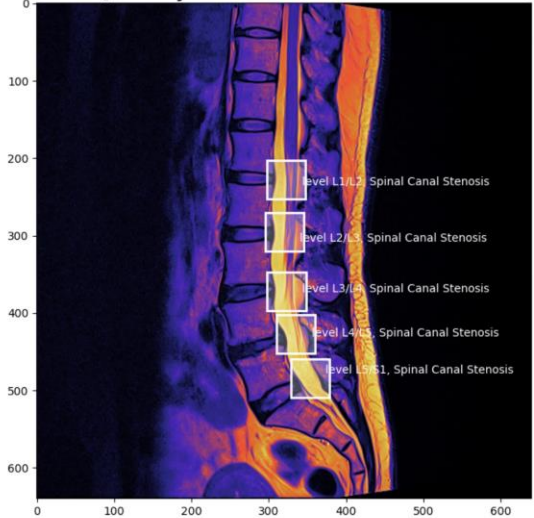
- Sagittal T2/STIR
- Sagittal T1
- Aixal T2

```
label_df[label_df['study_id']!=1002894806].reset_index(drop=True)
```

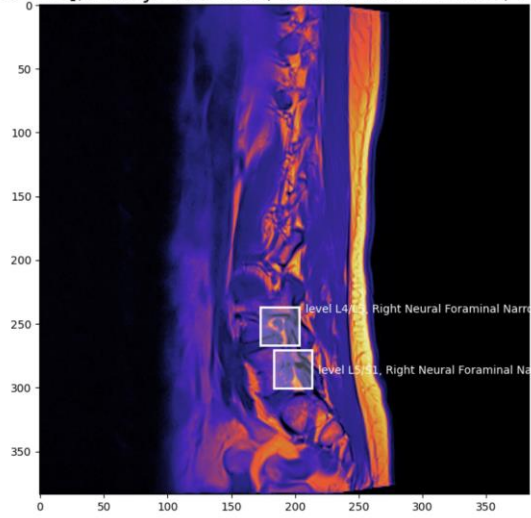
	study_id	condition	level	label	fold	series_id	instance_number	x	y	series_description
0	1002894806	Spinal Canal Stenosis	L1/L2	Normal/Mild	4	801316590	10	277.418838	129.058116	Sagittal T2/STIR
1	1002894806	Spinal Canal Stenosis	L2/L3	Normal/Mild	4	801316590	10	265.747495	181.130261	Sagittal T2/STIR
2	1002894806	Spinal Canal Stenosis	L3/L4	Normal/Mild	4	801316590	10	266.645291	240.384770	Sagittal T2/STIR
3	1002894806	Spinal Canal Stenosis	L4/L5	Moderate	4	801316590	10	278.316633	289.763527	Sagittal T2/STIR
4	1002894806	Spinal Canal Stenosis	L5/S1	Normal/Mild	4	801316590	10	287.294589	327.470942	Sagittal T2/STIR
5	1002894806	Left Neural Foraminal Narrowing	L1/L2	Normal/Mild	4	866293114	11	300.573643	136.589300	Sagittal T1
6	1002894806	Left Neural Foraminal Narrowing	L2/L3	Moderate	4	866293114	12	292.079763	178.228076	Sagittal T1
7	1002894806	Left Neural Foraminal Narrowing	L3/L4	Normal/Mild	4	866293114	12	287.830916	217.904097	Sagittal T1
8	1002894806	Left Neural Foraminal Narrowing	L4/L5	Normal/Mild	4	866293114	13	287.355258	262.140308	Sagittal T1
9	1002894806	Left Neural Foraminal Narrowing	L5/S1	Moderate	4	866293114	12	298.771054	301.144279	Sagittal T1
10	1002894806	Right Neural Foraminal Narrowing	L1/L2	Normal/Mild	4	866293114	4	303.106455	134.438321	Sagittal T1
11	1002894806	Right Neural Foraminal Narrowing	L2/L3	Normal/Mild	4	866293114	4	294.246810	178.106376	Sagittal T1
12	1002894806	Right Neural Foraminal Narrowing	L3/L4	Moderate	4	866293114	4	288.716012	221.649682	Sagittal T1
13	1002894806	Right Neural Foraminal Narrowing	L4/L5	Moderate	4	866293114	4	288.144718	263.039074	Sagittal T1
14	1002894806	Right Neural Foraminal Narrowing	L5/S1	Severe	4	866293114	4	294.813314	300.086831	Sagittal T1
15	1002894806	Left Subarticular Stenosis	L1/L2	Normal/Mild	4	1252873726	4	181.286671	164.198251	Axial T2
16	1002894806	Left Subarticular Stenosis	L2/L3	Normal/Mild	4	1252873726	12	178.021365	167.463557	Axial T2
17	1002894806	Left Subarticular Stenosis	L3/L4	Normal/Mild	4	1252873726	21	178.021365	163.265306	Axial T2
18	1002894806	Left Subarticular Stenosis	L4/L5	Moderate	4	1252873726	29	176.621948	156.734694	Axial T2
19	1002894806	Left Subarticular Stenosis	L5/S1	Normal/Mild	4	1252873726	38	182.219616	166.530612	Axial T2
20	1002894806	Right Subarticular Stenosis	L1/L2	Normal/Mild	4	1252873726	6	143.323615	167.930029	Axial T2
21	1002894806	Right Subarticular Stenosis	L2/L3	Normal/Mild	4	1252873726	13	144.723032	169.329446	Axial T2
22	1002894806	Right Subarticular Stenosis	L3/L4	Normal/Mild	4	1252873726	21	144.723032	164.664723	Axial T2
23	1002894806	Right Subarticular Stenosis	L4/L5	Severe	4	1252873726	29	142.857143	155.801749	Axial T2
24	1002894806	Right Subarticular Stenosis	L5/S1	Normal/Mild	4	1252873726	39	151.253644	162.332362	Axial T2

同一样本不同视角

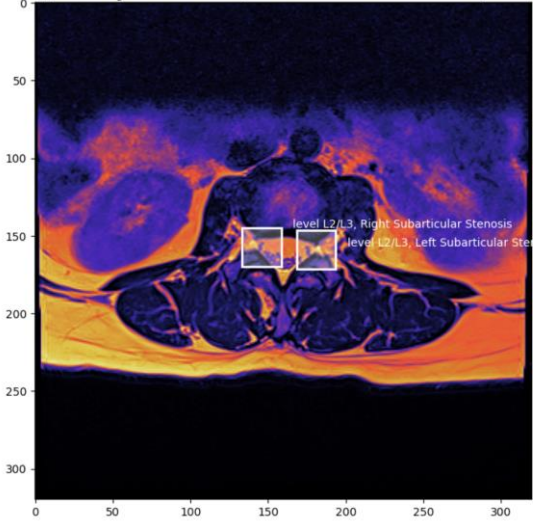
['Sagittal T2/STIR'], Study: 4003253, Series: 702807833, Instance: 8



['Sagittal T1'], Study: 4003253, Series: 1054713880, Instance: 4



['Axial T2'], Study: 4003253, Series: 2448190387, Instance: 11



提交格式与评价指标

```
In [1]: import pandas as pd
```

```
data = pd.read_csv('sample_submission.csv')
data
```

```
Out[1]:
```

	row_id	normal_mild	moderate	severe
0	44036939_left_neural_foraminal_narrowing_l1_l2	0.333333	0.333333	0.333333
1	44036939_left_neural_foraminal_narrowing_l2_l3	0.333333	0.333333	0.333333
2	44036939_left_neural_foraminal_narrowing_l3_l4	0.333333	0.333333	0.333333
3	44036939_left_neural_foraminal_narrowing_l4_l5	0.333333	0.333333	0.333333
4	44036939_left_neural_foraminal_narrowing_l5_s1	0.333333	0.333333	0.333333
5	44036939_left_subarticular_stenosis_l1_l2	0.333333	0.333333	0.333333
6	44036939_left_subarticular_stenosis_l2_l3	0.333333	0.333333	0.333333
7	44036939_left_subarticular_stenosis_l3_l4	0.333333	0.333333	0.333333
8	44036939_left_subarticular_stenosis_l4_l5	0.333333	0.333333	0.333333
9	44036939_left_subarticular_stenosis_l5_s1	0.333333	0.333333	0.333333
10	44036939_right_neural_foraminal_narrowing_l1_l2	0.333333	0.333333	0.333333
11	44036939_right_neural_foraminal_narrowing_l2_l3	0.333333	0.333333	0.333333
12	44036939_right_neural_foraminal_narrowing_l3_l4	0.333333	0.333333	0.333333
13	44036939_right_neural_foraminal_narrowing_l4_l5	0.333333	0.333333	0.333333
14	44036939_right_neural_foraminal_narrowing_l5_s1	0.333333	0.333333	0.333333
15	44036939_right_subarticular_stenosis_l1_l2	0.333333	0.333333	0.333333
16	44036939_right_subarticular_stenosis_l2_l3	0.333333	0.333333	0.333333
17	44036939_right_subarticular_stenosis_l3_l4	0.333333	0.333333	0.333333
18	44036939_right_subarticular_stenosis_l4_l5	0.333333	0.333333	0.333333
19	44036939_right_subarticular_stenosis_l5_s1	0.333333	0.333333	0.333333
20	44036939_spinal_canal_stenosis_l1_l2	0.333333	0.333333	0.333333
21	44036939_spinal_canal_stenosis_l2_l3	0.333333	0.333333	0.333333
22	44036939_spinal_canal_stenosis_l3_l4	0.333333	0.333333	0.333333
23	44036939_spinal_canal_stenosis_l4_l5	0.333333	0.333333	0.333333
24	44036939_spinal_canal_stenosis_l5_s1	0.333333	0.333333	0.333333

Evaluation

Submissions are evaluated using the average of sample weighted log losses and an `any_severe_spinal` prediction generated by the metric. The [metric notebook](#) can be found [here](#).

The sample weights are as follows:

- 1 for normal/mild.
- 2 for moderate.
- 4 for severe.

For each row ID in the test set, you must predict a probability for each of the different severity levels. The file should contain a header and have the following format:

```
row_id,normal_mild,moderate,severe
123456_left_neural_foraminal_narrowing_l1_l2,0.333,0.333,0.333
123456_left_neural_foraminal_narrowing_l2_l3,0.333,0.333,0.333
123456_left_neural_foraminal_narrowing_l3_l4,0.333,0.333,0.333
etc.
```

In rare cases the lowest vertebrae aren't visible in the imagery. You still need to make predictions (nulls will cause errors), but those rows will not be scored.

For this competition, the `any_severe_scalar` has been set to `1.0`.

算法思路

算法基本流程

本次比赛利用yolov8识别和分类腰椎退化症状，算法流程：

1. 数据处理：根据提供的磁共振成像图片以及腰椎椎体的定点坐标位置信息构建目标检测数据，在多种放缩尺寸图像上设置合适大小的box，最大化保留严重退化（Severe）特征信息。
2. 建模1：将病例图像分成三大类：
 - spinal canal stenosis 椎管狭窄（15类 1*5*3）
 - subarticular stenosis 椎间孔狭窄（30类 2*5*3）
 - neural foraminal narrowing 神经根管狭窄（30类 2*5*3）

进行建立模型训练。在训练时对图像进行增强。

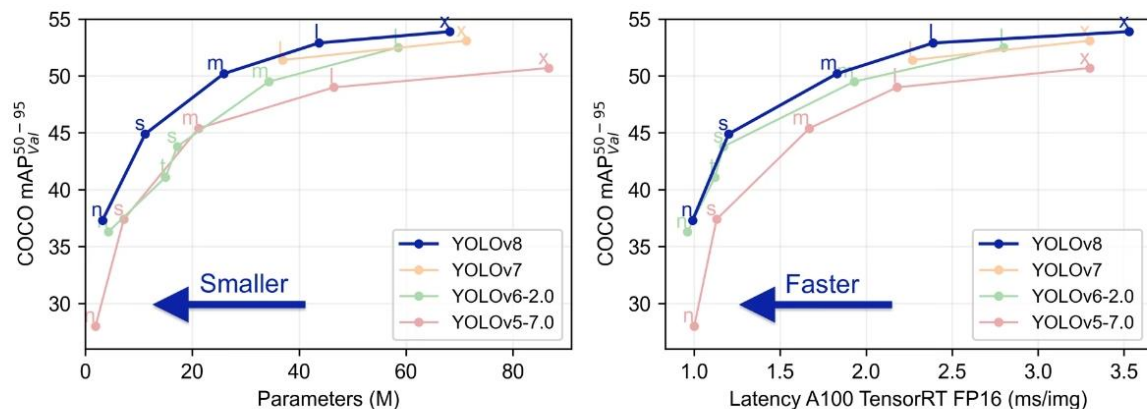
3. 建模2：此方式与建模1类似，不同之处在于标签屏蔽。由于存在单个病例图像具有多个标签的情况，而且yolov8不具有多标签多分类功能，因此在数据处理阶段将具有多标签（同时存在严重退化与非严重退化）的图片去除非严重退化（即normal_mild和moderate）类别，加强对严重退化类别的识别能力。
4. 推理：在每个病例的三大类图像中，分别采取对应的模型推理，在每个系列图像中分别得到所有类型的概率，取每种类别的最大值概率即为单个病例每种类别的概率，将所得概率归一化即为最后分类预测结果。

算法思路

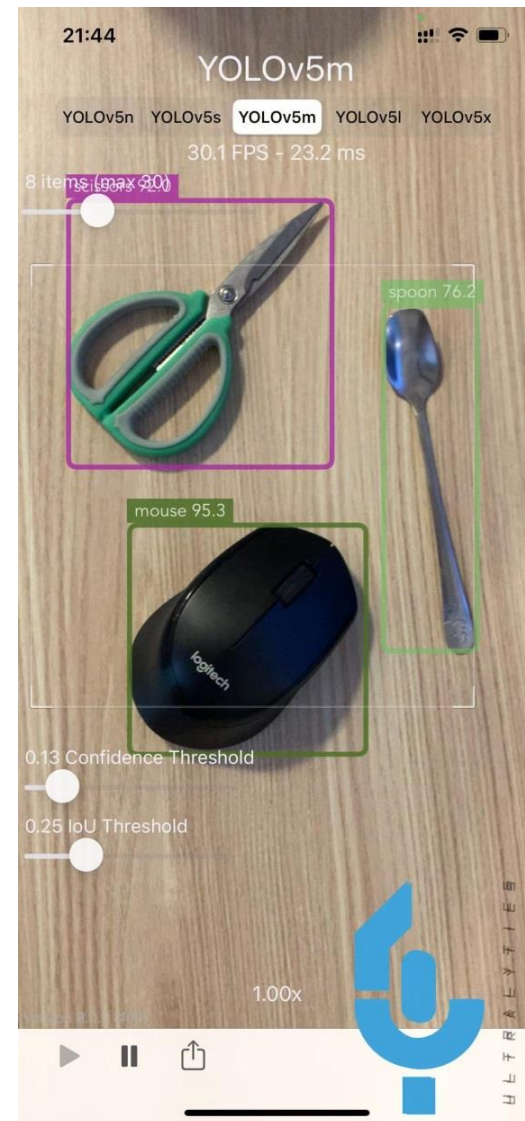
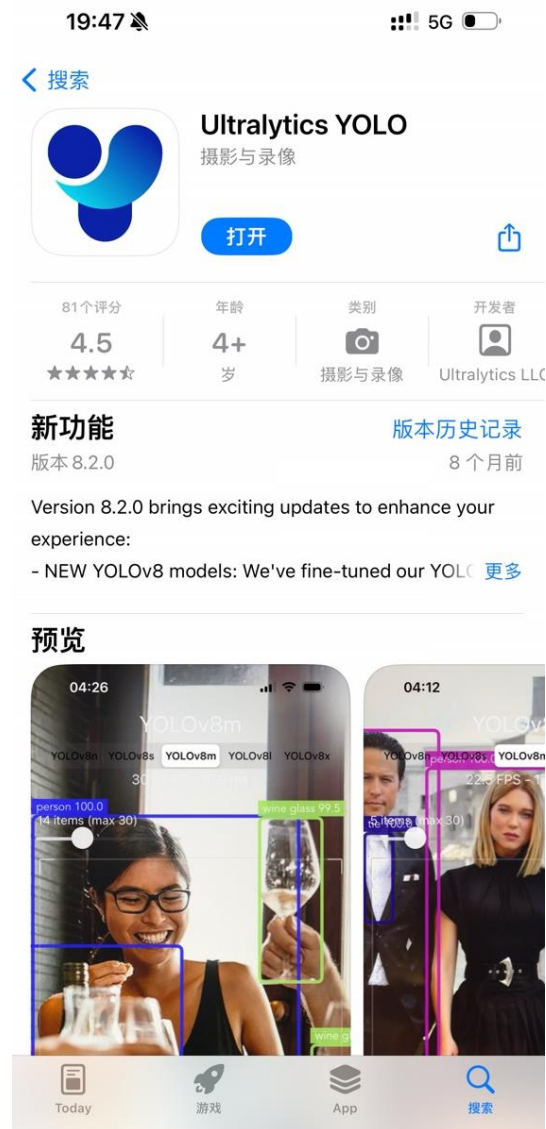
模型简介: YOLOv8

Overview

YOLOv8 is the latest iteration in the YOLO series of real-time object detectors, offering cutting-edge performance in terms of accuracy and speed. Building upon the advancements of previous YOLO versions, YOLOv8 introduces new features and optimizations that make it an ideal choice for various [object detection](#) tasks in a wide range of applications.

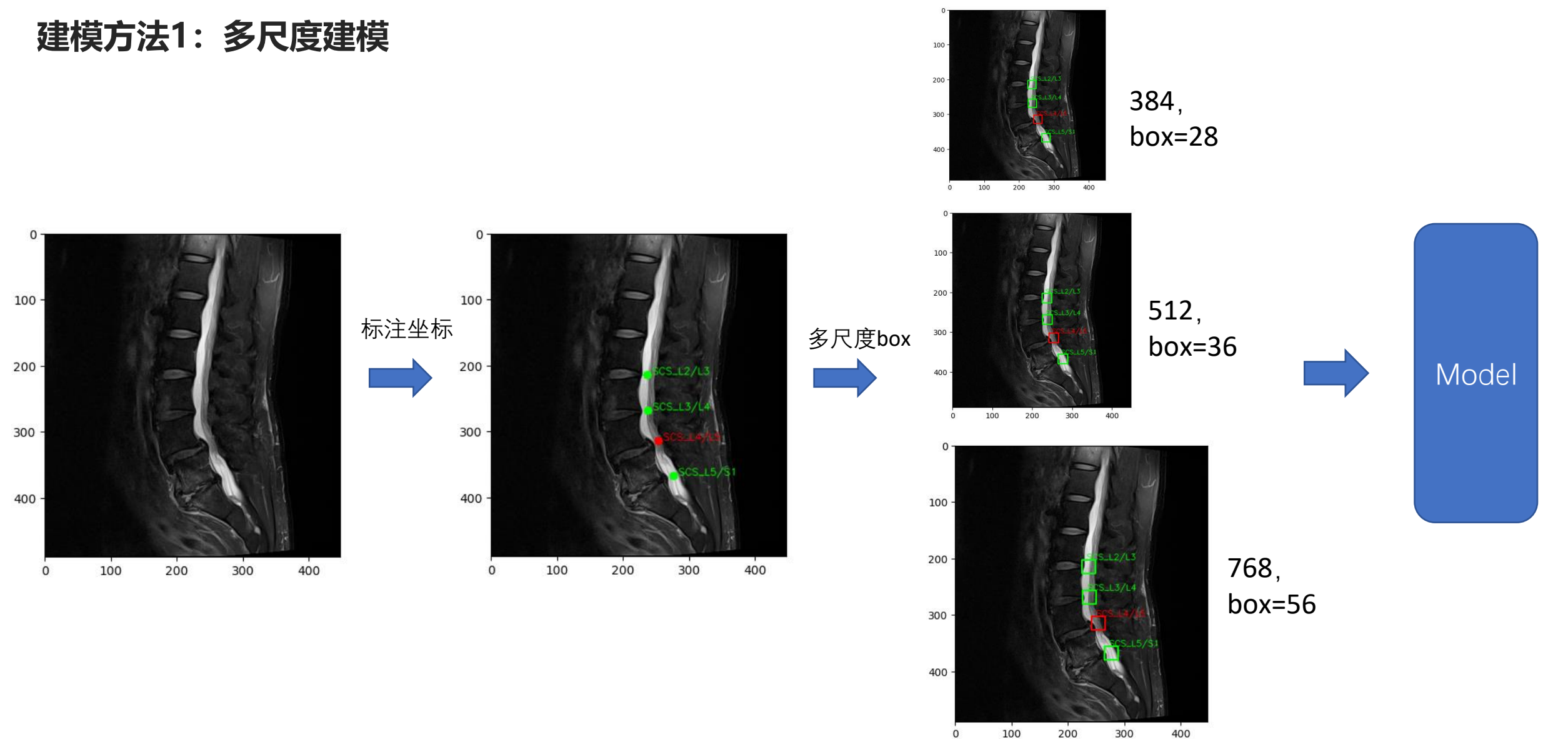


<https://docs.ultralytics.com/models/yolov8/>



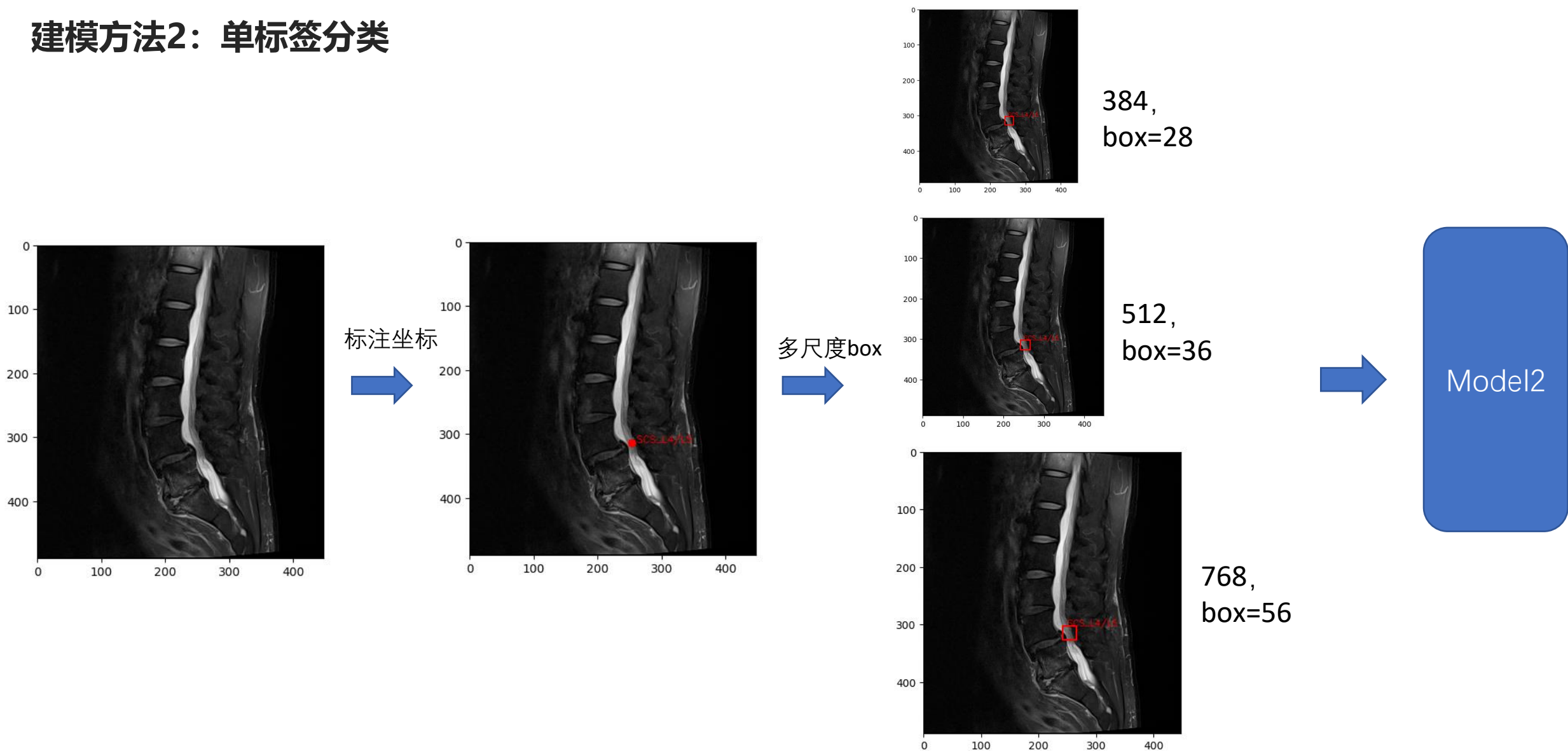
算法思路

建模方法1：多尺度建模

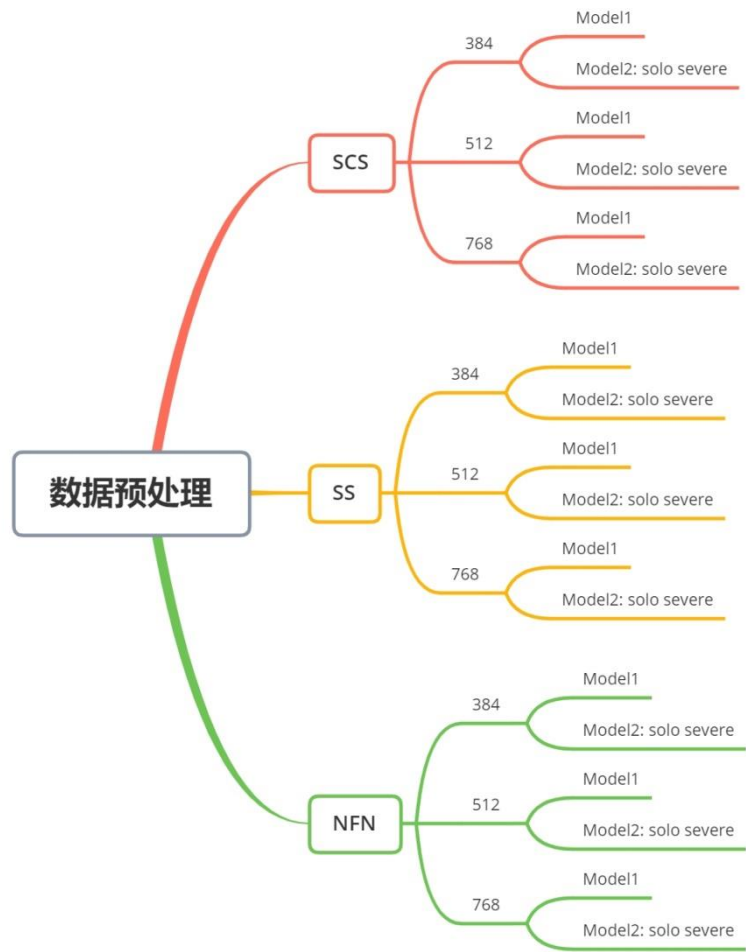


算法思路

建模方法2：单标签分类



算法流程图



Yolo训练配置文件

```
path: /root/data/rsna/data/lsdc-yolo-data-384/nfn_data_fold0/data_fold0
train: images/train
val: images/val
test: images/val
#Classes
names:
...0: left_neural_foraminal_narrowing_l1_l2_normal/mild
...1: left_neural_foraminal_narrowing_l1_l2_moderate
...2: left_neural_foraminal_narrowing_l1_l2_severe
...3: left_neural_foraminal_narrowing_l2_l3_normal/mild
...4: left_neural_foraminal_narrowing_l2_l3_moderate
...5: left_neural_foraminal_narrowing_l2_l3_severe
...6: left_neural_foraminal_narrowing_l3_l4_normal/mild
...7: left_neural_foraminal_narrowing_l3_l4_moderate
...8: left_neural_foraminal_narrowing_l3_l4_severe
...9: left_neural_foraminal_narrowing_l4_l5_normal/mild
...10: left_neural_foraminal_narrowing_l4_l5_moderate
...11: left_neural_foraminal_narrowing_l4_l5_severe
...12: left_neural_foraminal_narrowing_l5_s1_normal/mild
...13: left_neural_foraminal_narrowing_l5_s1_moderate
...14: left_neural_foraminal_narrowing_l5_s1_severe
...15: right_neural_foraminal_narrowing_l1_l2_normal/mild
...16: right_neural_foraminal_narrowing_l1_l2_moderate
...17: right_neural_foraminal_narrowing_l1_l2_severe
...18: right_neural_foraminal_narrowing_l2_l3_normal/mild
...19: right_neural_foraminal_narrowing_l2_l3_moderate
...20: right_neural_foraminal_narrowing_l2_l3_severe
...21: right_neural_foraminal_narrowing_l3_l4_normal/mild
...22: right_neural_foraminal_narrowing_l3_l4_moderate
...23: right_neural_foraminal_narrowing_l3_l4_severe
...24: right_neural_foraminal_narrowing_l4_l5_normal/mild
...25: right_neural_foraminal_narrowing_l4_l5_moderate
...26: right_neural_foraminal_narrowing_l4_l5_severe
...27: right_neural_foraminal_narrowing_l5_s1_normal/mild
...28: right_neural_foraminal_narrowing_l5_s1_moderate
...29: right_neural_foraminal_narrowing_l5_s1_severe
```

```
path: /root/data/rsna/data/yolo-data-512-severe/scs_data
train: images/train
val: images/val
test: images/val
#Classes
names:
...0: spinal_canal_stenosis_l1_l2_normal/mild
...1: spinal_canal_stenosis_l1_l2_moderate
...2: spinal_canal_stenosis_l1_l2_severe
...3: spinal_canal_stenosis_l2_l3_normal/mild
...4: spinal_canal_stenosis_l2_l3_moderate
...5: spinal_canal_stenosis_l2_l3_severe
...6: spinal_canal_stenosis_l3_l4_normal/mild
...7: spinal_canal_stenosis_l3_l4_moderate
...8: spinal_canal_stenosis_l3_l4_severe
...9: spinal_canal_stenosis_l4_l5_normal/mild
...10: spinal_canal_stenosis_l4_l5_moderate
...11: spinal_canal_stenosis_l4_l5_severe
...12: spinal_canal_stenosis_l5_s1_normal/mild
...13: spinal_canal_stenosis_l5_s1_moderate
...14: spinal_canal_stenosis_l5_s1_severe
```


算法思路：后处理

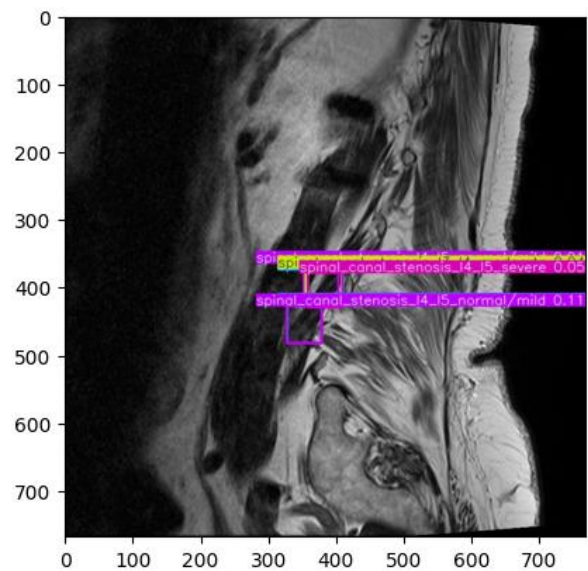


image1 predict:
class:[9, 11, 10, 11, 7, 9, 10]
prob:[0.1131, 0.1010, 0.0661,
0.0503, 0.0412, 0.0135, 0.0119]

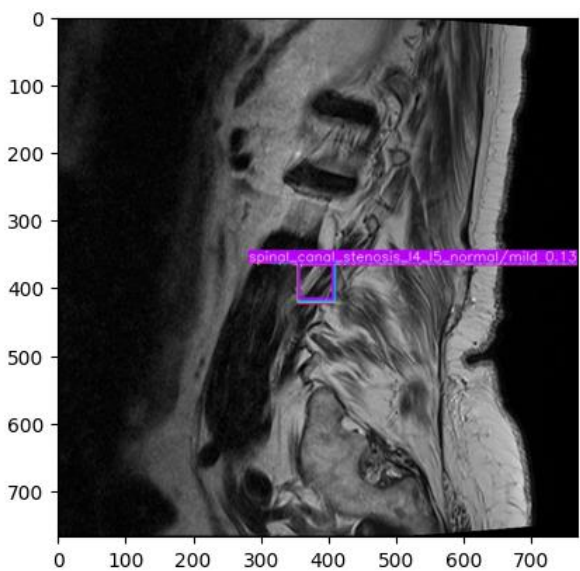


image2 predict:
class:[9, 10, 7, 11, 6]
prob:[0.1294, 0.1028, 0.0792, 0.0410, 0.0356]

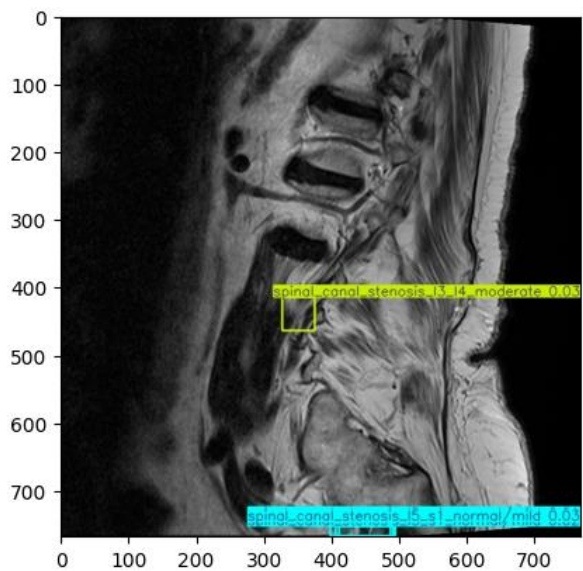
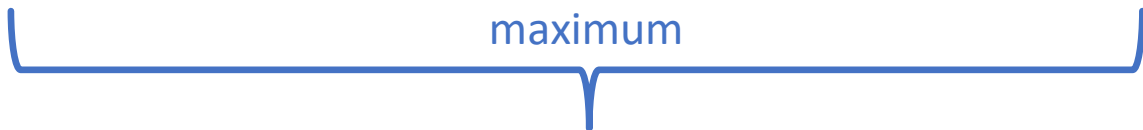


image3 predict:
class:[12, 7, 12, 12]
prob:[0.0341, 0.0264, 0.0209, 0.0141]

{0: 'spinal_canal_stenosis_l1_l2_normal/mild',
1: 'spinal_canal_stenosis_l1_l2_moderate',
2: 'spinal_canal_stenosis_l1_l2_severe',
3: 'spinal_canal_stenosis_l2_l3_normal/mild',
4: 'spinal_canal_stenosis_l2_l3_moderate',
5: 'spinal_canal_stenosis_l2_l3_severe',
6: 'spinal_canal_stenosis_l3_l4_normal/mild',
7: 'spinal_canal_stenosis_l3_l4_moderate',
8: 'spinal_canal_stenosis_l3_l4_severe',
9: 'spinal_canal_stenosis_l4_l5_normal/mild',
10: 'spinal_canal_stenosis_l4_l5_moderate',
11: 'spinal_canal_stenosis_l4_l5_severe',
12: 'spinal_canal_stenosis_l5_s1_normal/mild',
13: 'spinal_canal_stenosis_l5_s1_moderate',
14: 'spinal_canal_stenosis_l5_s1_severe'}



class:[6, 7, 9, 10, 11, 12]
prob:[0.0356, 0.0792, 0.1294, 0.1028, 0.1010, 0.0341]

Normalize



class:[..., 9, 10, 11, ...]
prob:[..., 0.3884, 0.3085, 0.3031, ...]

算法思路

算法亮点

1. 根据图像与坐标信息使用目标检测模型作分类训练。
2. 多尺度修改图像获取丰富图像特征训练模型。
3. 修改目标检测模型的推理过程得到分类概率。
4. 集成多个模型结果提高预测泛化。